

# CYBERSECURITY INCIDENT INVESTIGATION REPORT

## **Phishing-Led Malware Investigation Dai Wok Foods**

*Professional Portfolio Case Study*

*Prepared from documented security investigation evidence*

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## 1. Executive Summary

This investigation examined a suspected phishing-led security incident affecting Dai Wok Foods. The available evidence identified an external email campaign targeting company personnel, followed by interaction with a malicious link hosted on `complaints-cityofficialsfood.com`. The investigation then correlated email telemetry, employee records, passive DNS data, authentication activity, file creation events, and process execution records to reconstruct the observed attack sequence.

A key affected employee identified during the investigation was John Garcia, associated with the hostname `LVJW-LAPTOP`. Evidence documented that he interacted with the phishing link on `2023-04-03T18:04:56Z`. Subsequent investigation identified the downloaded file `large_order.xlsx` and suspicious command execution on the affected endpoint. A PowerShell command was observed with non-interactive execution, profile suppression, execution-policy bypass, and a referenced script named `c5k3fsys.3bp.ps1`.

The documented investigation further linked the observed malware activity to a PowerShell-based Trojan classification and recorded ransomware as the resulting malware category. The findings demonstrate an end-to-end analytical workflow from initial phishing identification through infrastructure analysis and endpoint investigation.

## 2. Investigation Scope

The investigation focused on determining:

- The identity of targeted employees and relevant corporate assets.
- The characteristics and origin of the suspicious email campaign.
- Whether users interacted with the malicious infrastructure.
- The infrastructure associated with the phishing domain.
- The endpoint and file activity following the user interaction.
- Whether suspicious process execution indicated malware activity.
- The likely nature of the malware and its operational impact.

## 3. Methodology and Evidence Sources

The investigation used correlation across multiple telemetry sources documented in the challenge environment:

- Employees — user identity, email address, IP address, hostname, and role correlation.
- Email — sender, recipient, subject, reply-to relationships, and malicious links.
- SecurityAlerts — identification of alerts associated with the targeted user or device.
- PassiveDns — historical domain-to-IP infrastructure analysis.
- InboundNetworkEvents — investigation of activity originating from suspicious infrastructure.
- AuthenticationEvents — identification of successful authentication activity from unusual source IP addresses.
- FileCreationEvents — identification and tracking of downloaded or malicious files.
- ProcessEvents — reconstruction of suspicious command and script execution.

## 4. Key Findings

### 4.1 Initial Phishing Activity

The investigation identified Delphia Evans as an employee of Dai Wok Foods with the username deevans and hostname ABVJ-MACHINE. Email analysis identified a suspicious message with the subject "[EXTERNAL] Formal action on food poisoning." The documented investigation identified county.county@yahoo.com as a suspect sender associated with this activity.

| Attribute          | Observed Value                             |
|--------------------|--------------------------------------------|
| Targeted employee  | Delphia Evans                              |
| Username           | deevans                                    |
| Hostname           | ABVJ-MACHINE                               |
| Suspicious subject | [EXTERNAL] Formal action on food poisoning |
| Suspect sender     | county.county@yahoo.com                    |
| Recorded timestamp | 2023-04-02T20:40:54Z                       |

### 4.2 Malicious Link and User Interaction

The phishing investigation identified the domain complaints-cityofficialsfood.com as malicious infrastructure. A suspicious email associated with official@verizon.com contained a link leading to the domain, with the documented file path ending in large\_order.xlsx.

The investigation recorded that John Garcia clicked the phishing link on 2023-04-03T18:04:56Z. John Garcia was associated with the username jogarcia, email john\_garcia@daiwokfoods.com, IP address 192.168.3.86, and hostname LVJW-LAPTOP.

| Indicator                 | Value                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------|
| Malicious domain          | complaints-cityofficialsfood.com                                                         |
| Full observed link        | https://complaints-cityofficialsfood.com/search/published/images/images/large_order.xlsx |
| Affected employee         | John Garcia                                                                              |
| Hostname                  | LVJW-LAPTOP                                                                              |
| Username                  | jogarcia                                                                                 |
| Recorded link interaction | 2023-04-03T18:04:56Z                                                                     |

### 4.3 Threat Infrastructure Analysis

Passive DNS analysis identified nine historical records for complaints-cityofficialsfood.com. The documented infrastructure included multiple IP addresses over time. After the recorded user interaction, the investigation identified an association between the domain and 179.58.169.157 on 2023-04-04T18:32:25Z. The notes further recorded that this IP originated from a server in Bolivia.

| Timestamp            | Observed IP     |
|----------------------|-----------------|
| 2023-03-26T13:29:46Z | 134.173.137.45  |
| 2023-03-28T17:05:34Z | 177.43.27.137   |
| 2023-03-30T14:03:02Z | 188.216.230.252 |
| 2023-03-31T04:23:15Z | 131.162.160.193 |
| 2023-04-04T18:32:25Z | 179.58.169.157  |
| 2023-04-15T11:10:23Z | 179.58.169.157  |
| 2023-04-17T02:21:38Z | 177.43.27.137   |
| 2023-04-18T03:51:38Z | 77.112.237.13   |

|                      |                |
|----------------------|----------------|
| 2023-04-25T12:16:50Z | 189.100.73.140 |
|----------------------|----------------|

## 4.4 File and Endpoint Investigation

The malicious link led investigators to the filename `large_order.xlsx`. File creation analysis identified the file at the following documented location on the affected system:

`C:\Users\jogarcia\Downloads\large_order.xlsx`

From the investigation, it was determined that the file was not flagged as suspicious by VirusTotal at the time of the documented analysis. This illustrates the importance of behavioral and telemetry-based investigation rather than relying exclusively on file reputation.

## 4.5 Suspicious Process Execution

Process telemetry from LVJW-LAPTOP showed suspicious command execution after the relevant incident period. A significant command line launched PowerShell through `cmd.exe` and included non-interactive execution, profile suppression, execution-policy bypass, and execution of a script from an administrative share.

```
cmd.exe /c start %SYSTEMROOT%\system32\WindowsPowerShell\v1.0\powershell.exe -noni -nop -
exe bypass -f \\share1\Admin$\c5k3fsys.3bp.ps1
```

The associated SHA-256 value documented in the investigation was:

662124b0c998fd0826c192514b1f57f8002f2ab031996aa6dd7832f561679779

## 4.6 Malware Assessment

The documented malware classification identified the activity as a PowerShell-based Trojan (trojan.powershell/pshell). The investigation notes further recorded ransomware as the malware category associated with the assessed incident. This conclusion was based on the evidence and external malware identification steps documented during the challenge.

## 5. Reconstructed Incident Timeline

| Time                          | Event                                                                           |
|-------------------------------|---------------------------------------------------------------------------------|
| 2023-04-02T20:40:54Z          | Suspect phishing email activity recorded from county.county@yahoo.com.          |
| 2023-04-03T18:04:56Z          | John Garcia recorded as clicking the phishing link.                             |
| After link interaction        | <code>large_order.xlsx</code> identified in the investigation workflow.         |
| Endpoint investigation period | File identified at <code>C:\Users\jogarcia\Downloads\large_order.xlsx</code> .  |
| Post-download activity        | Suspicious PowerShell execution observed on LVJW-LAPTOP.                        |
| 2023-04-04T18:32:25Z          | complaints-cityofficialsfood.com associated with 179.58.169.157 in Passive DNS. |

## 6. Attack Chain Assessment

| Stage              | Assessment                                                                           |
|--------------------|--------------------------------------------------------------------------------------|
| Initial Access     | External phishing emails targeting Dai Wok Foods employees.                          |
| Social Engineering | Food-poisoning themed message designed to encourage recipient interaction.           |
| User Execution     | John Garcia documented as clicking the malicious link.                               |
| Malicious Content  | large_order.xlsx delivered through attacker-controlled infrastructure.               |
| Execution          | Suspicious PowerShell command launched through cmd.exe with execution-policy bypass. |
| Malware            | PowerShell-based Trojan classification documented during analysis.                   |
| Impact             | Ransomware recorded as the assessed malware category.                                |

## 7. Indicators of Compromise

| Type          | Indicator                                                                                |
|---------------|------------------------------------------------------------------------------------------|
| Domain        | complaints-cityofficialsfood.com                                                         |
| URL           | https://complaints-cityofficialsfood.com/search/published/images/images/large_order.xlsx |
| File          | large_order.xlsx                                                                         |
| Script        | c5k3fsys.3bp.ps1                                                                         |
| SHA-256       | 662124b0c998fd0826c192514b1f57f8002f2ab031996aa6dd7832f561679779                         |
| IP            | 179.58.169.157                                                                           |
| Email         | county.county@yahoo.com                                                                  |
| Email         | official@verizon.com                                                                     |
| Affected host | LVJW-LAPTOP                                                                              |

## 8. Analytical Conclusions

The available evidence supports a coherent phishing-led intrusion sequence. The investigation established a connection between suspicious external emails, attacker-controlled web infrastructure, user interaction, a downloaded file, and suspicious PowerShell execution on an identified employee endpoint.

A notable investigative lesson from the case is that the documented downloaded file was not flagged as suspicious by VirusTotal, while subsequent endpoint telemetry provided stronger behavioral evidence of malicious activity. This reinforces the value of cross-source correlation in incident response.

Based on the documented evidence, the incident should be treated as a significant endpoint compromise with phishing as the apparent initial delivery mechanism and PowerShell execution as a critical post-click

indicator. The documented malware assessment ultimately associated the activity with a PowerShell-based Trojan and ransomware.

## 9. Recommended Response Actions

- Isolate affected endpoints and preserve volatile and endpoint telemetry for further forensic review.
- Block the identified malicious domain, URLs, and associated indicators across DNS, proxy, email, and endpoint controls.
- Search the environment for the identified filename, script name, hash, and suspicious PowerShell command patterns.
- Review authentication activity for affected users and reset credentials where compromise cannot be ruled out.
- Perform enterprise-wide hunting for connections to the identified attacker infrastructure.
- Review PowerShell logging and strengthen alerting for execution-policy bypass and suspicious remote script execution.
- Conduct targeted phishing awareness reinforcement using the observed social-engineering pattern.
- Validate backup and ransomware recovery procedures following any confirmed encryption or destructive activity.

## 10. Analyst Skills Demonstrated

- Security incident investigation and timeline reconstruction
- KQL-based log analysis and cross-table correlation
- Phishing and malicious email investigation
- DNS and attacker infrastructure analysis
- Endpoint and file telemetry analysis
- Process and PowerShell execution analysis
- Indicator of Compromise identification
- Threat hunting and evidence correlation
- Malware triage and behavioral assessment
- Incident reporting and technical communication

## 11. Portfolio Note

This report is a portfolio case study reconstructed from a documented cybersecurity investigation challenge on Kc7 Cybergame platform. The organization name, users, hosts, domains, IP addresses, and events reflect the challenge dataset and are presented to demonstrate the investigative methodology, analytical reasoning, and incident-response skills I used during the investigation of this challenge.